

Problem 10.36

An investment is made in an account in which the amount function is given by

$$A(t) = t^2 + t + 2.25, \quad 0 \leq t \leq 4.$$

Compute the ratio

$$\frac{i_3}{\delta_2}.$$

Solution

The effective interest rate during the third period is

$$i_3 = \frac{A(3) - A(2)}{A(2)}.$$

The relevant account values are

$$A(3) = 3^2 + 3 + 2.25 = 14.25 = \frac{57}{4},$$

$$A(2) = 2^2 + 2 + 2.25 = 8.25 = \frac{33}{4}.$$

Therefore,

$$\begin{aligned} i_3 &= \frac{14.25 - 8.25}{8.25} \\ &= \frac{6}{8.25} \\ &= \frac{8}{11}. \end{aligned}$$

The force of interest at time t is

$$\delta_t = \frac{A'(t)}{A(t)}.$$

Since

$$A'(t) = 2t + 1,$$

we have

$$\begin{aligned} \delta_2 &= \frac{2(2) + 1}{2^2 + 2 + 2.25} \\ &= \frac{5}{8.25} \\ &= \frac{20}{33}. \end{aligned}$$

Consequently,

$$\begin{aligned} \frac{i_3}{\delta_2} &= \frac{\frac{8}{11}}{\frac{20}{33}} \\ &= \frac{8}{11} \cdot \frac{33}{20} \\ &= \frac{6}{5}. \end{aligned}$$

Thus,

$$\frac{i_3}{\delta_2} = \frac{6}{5} = 1.2.$$